

Paul Walko

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WORK EXPERIENCE

Senior DevOps Engineer

Ozmo, Inc

August 2019 - Present | Blacksburg, VA

- **AWS Architecture:** Architect production applications (ECS, Fargate, Lambda, RDS, OpenSearch) for maintainability and security; lead quarterly DevOps planning as most senior DevOps Engineer
- **AI:** Experienced with using Claude Code to quickly troubleshoot AWS, all from the CLI; Adapted existing CI/CD pipelines to be more “AI Friendly” allowing other engineers to have less friction when deploying
- **SOC 2:** Own end-to-end compliance program: IAM access reviews, vulnerability management, CI/change controls, audit logging, encryption, backup/DR testing, vendor risk reviews; liaison with external auditor
- **Observability:** Built ELK + Grafana stack ingesting 30M+ log events/day; primary system for production alerting and incident response
- **CI/CD:** Led migration efforts from Jenkins to GitHub Actions and frequently assist other engineers with CI/CD related tasks

Systems Engineer

Solers, Inc

January 2019 - July 2019 | Arlington, VA

- Worked with Docker Swarm and Ansible
- Deployed central logging solution for clustered environment

Cyber Security Engineer

Cisco

May 2017 - August 2018 | Raleigh-Durham, NC

Research Assistant

Hume Center at Virginia Tech

September 2016 - December 2017 | Blacksburg, VA

SKILLS

AWS: API Gateway, CloudFormation, CloudFront, CloudWatch, EC2, ECS, Fargate, Elasticsearch, OpenSearch, Lambda, RDS, Route53, S3

GIS Tools: QGIS, GDAL, Leaflet, GeoServer, MapProxy

Languages: Bash, C, Java, JavaScript, Node.js, Python, Ruby

Tools: Ansible, AWS CDK, Docker, GitHub Actions, Google Workspace, Git, Grafana, Helm, Jenkins, Kubernetes, Librato, Linux, New Relic, Tailscale, Ubiquiti

Other: Amateur Radio (KN4VDC), Argentine Tango (assist with lessons)

EDUCATION

Virginia Tech

Bachelor of Science in Computer Science, Minor in Cyber Security

2015 - 2018 | Blacksburg, VA

PROJECTS

Self-Hosted Services

- **Kubernetes Cluster:** IPv6-only multi-node k8s cluster on Tailscale running Browsertrix, Forgejo, Immich, Nextcloud, RustFS
- **Cavepedia:** github.com/cavepedia/cavepediav2. Semantic search + chat over decades of NSS caving documents in partnership with the National Speleological Society; pgvector embeddings, FastMCP server, PydanticAI agent, Auth0 RBAC.
- **Mergin Maps:** maps.caving.dev. Used on remote caving expeditions to manage GIS.
- **Mastodon:** social.bigcavemaps.com
- **Geoserver/Leaflet/MapProxy:** Used to analyze lidar and find potential caves.

FOSDEM 2026 Air Quality Monitoring

- Co-built distributed CO₂/temp/humidity sensor network deployed across 20+ devrooms (Meshtastic mesh, MQTT, Postgres, Grafana); results shared with FOSDEM staff for ventilation planning. paul.walko.org/blog/2026-02-07_fosdem-part2

Free/Open Software Contributions

- **OpenStreetMap:** Contributor since 2013, especially not well known trails.
- **Archive Team:** Help archive Internet services and manage massive amounts of data; contributor since 2019.

ORGANIZATIONS

VPI Cave Club | vpicave.club (Student Grotto of the NSS, member since 2020)

- **Survey/Cartography:** Survey and map unexplored cave passage, including data entry and processing to create a finished map.
- **Training:** Teach new members basic caving skills, including vertical competence when climbing or abseiling rope.
- **Events:** Help organize regional caving events with 200+ attendees.

Virginia Tech Linux/Unix Users Group (VTLUUG) | vtluug.org, github.com/vtluug (member since 2016)

- **FreeIPA/LDAP:** Maintain accounts for email, shell access, and webserver.
- **Rebuilt Infrastructure:** Updated services to deploy with Ansible and run all services with Docker.
- **Organized funds:** Obtained \$1300 for new servers and equipment; oversaw purchasing additional components.

Outdoor Club at Virginia Tech (OCVT) | ocvt.club, github.com/ocvt (member since 2017)

- **Dolabra:** Built backend API in Golang and SQLite + front-end in plain HTML/JavaScript. Used OpenID Connect for auth, Stripe for payments, and Docker for testing and deployment in AWS.
- **Appalachian Trail Maintenance:** Regularly clear trail of trees, brush, and clean shelters.

TALKS

Cave Rescue Communications (Toorcamp 2024): archive.is/iX4s8

Austrian Expedition to the Loser Plateau (NSS Convention 2023)

Newberry's Cave Exploration (NSS Convention 2022)